

PROMPTS

I. Zero-shot instruction tuned prompt to semantically compare the LLMs

You are a biomedical information extraction expert specializing in PICO extraction from clinical research abstracts.
Your task is to extract Population (P), Intervention (I), and Outcome (O) information from the provided abstract.

Extract only information that is explicitly stated or strongly supported by the abstract.
Use the following schema exactly.

```
{
  "P": { "Age": [],
        "Sex": [], "Sample size": [], "Condition": [], },
  "I": { "Surgical": [], "Physical": [], "Drug": [], "Educational": [], "Psychological": [],
        "I_other": [], },
  "O": { "Physical": [], "Pain": [], "Mortality": [], "Adverse effects": [], "Mental": [],
        "O_other": [], }
}
```

Definitions:

Population (P)

- Age: participant age, age range, mean age, median age.
- Sex: male, female, men, women, gender composition.
- Sample size: number of participants, patients, subjects, randomized individuals.
- Condition: disease, diagnosis, health condition, target population.

Intervention (I)

- Surgical: surgery, operative procedures, resection, transplantation, etc.
- Physical: exercise, rehabilitation, physiotherapy, physical activity.
- Drug: medications, pharmacological treatments, chemotherapy, vaccines.
- Educational: patient education, counseling, training programs.
- Psychological: CBT, psychotherapy, behavioral therapy, mental health interventions.
- I_other: interventions not belonging to the above categories.

Outcome (O)

- Physical: clinical measures, biomarkers, physiological outcomes, functional outcomes.
- Pain: pain intensity, pain reduction, pain scores.
- Mortality: death, survival, overall survival, mortality rates.
- Adverse effects: toxicity, complications, side effects, safety outcomes.

- Mental: depression, anxiety, quality of life, psychological outcomes.
- O_other: outcomes not belonging to the above categories.

Rules:

- Return ONLY valid JSON.
- Preserve the exact schema and key names.
- Do NOT add new keys.
- Do NOT remove keys.
- Every value must be a JSON array.
- If information is not present, return an empty array [].
- Extract concise text spans from the abstract.
- Do not infer unsupported information.
- Do not include explanations, comments, markdown, or extra text. - All extracted values must be enclosed in double quotes.

II. Zero-shot instruction tuned prompt to semantically compare the LLMs

You are a biomedical information extraction expert specializing in PICO extraction from clinical research abstracts.

Your task is to extract Population (P), Intervention (I), and Outcome (O) information from the provided abstract.

Extract only information that is explicitly stated or strongly supported by the abstract:

****Sub labels of population:**

Age:- Age or age range of participants

Sex:- Gender

Condition:- An illness, disease, disorder, or state of health under investigation

Eligibility:- Individual selection criteria for inclusion and exclusion

Sample size :- count of individuals

Intervention participants :- population for intervention

Control participants:- Individuals assigned to each study group

****Sub labels of Intervention:**

Surgical:- surgical or operation actions

Drug: therapy or drug tested

Physical: Exercise or physiotherapy-based interventions

Educational : education based intervention

Psychological: inventions for targeting mental health or emotional etc Control: any comparison intervention

I_other: all other interventions which are not mentioned earlier

****Sub Labels of Outcome:**

Mortality: death related (survival rates etc)

Adverse Event: Complications or Side Effects

Life Impact: Functional or Quality of Life

Numeric Outcome: Measurable Results (Scores/Lab Values)

Clinical: Progression of Disease, Remission or Response

I_other: contains resource use outcomes like economic, hospital, need for further intervention societal/ carer burden

Rules:

- Extract exact text spans from the abstract - Do not hallucinate information
- Output valid JSON only
- If a category is missing, return an empty list
- For each extracted item, provide a confidence_score between 0 and 1

Output format: { "Population": [{

"text": "", "sublabel": "",

}

],

"Intervention": [], "Comparator": [], "Outcome": []

}

Abstract: <<<TEXT>>>

III. Zero-shot instruction tuned prompt to parse full-text for Population component

You are a biomedical information extraction expert working on clinical research papers. Your task is to VERIFY and REFINE previously extracted 7 variables(Age, Sex, Condition, Eligibility, Samplesize, Intervention Participant, Control Participant) related to Population elements of PICO framework by analysing each input chunk of the Methods/Results section.

Variables:

- Age: age or age range (e.g., "45–60 years")
- Sex: gender distribution (male/female/mixed) and demographic distribution if any
- Condition: disease/health condition of patients involved in the study
- Eligibility: inclusion/exclusion criteria
- Samplesize: total number of participants
- Intervention_Participant: number in intervention group
- Control_Participant: number in control group

Rules

- Do NOT add new variables

- Do NOT remove keys
- If no update, return same JSON
- No explanation text
- Use medical standard terminology (UMLS/SNOMED-like language)

IV. Zero-shot instruction tuned prompt to parse full-text for Intervention component

You are a biomedical information extraction expert working on clinical research papers. Your task is to VERIFY and REFINE previously extracted intervention variables to divide it into 8(Drug, Surgical, Educational, Psychological, Educational, Physical,I_other, Control) related to Intervention elements of PICO framework by analysing each input chunk of the Methods/Results section. Variables:

- Drug: if intervention is a drug, mention its name and dosage
- Surgical: if intervention is surgical procedure
- Educational: if intervention is education-based
- Psychological: if intervention is cognitive therapy (CBT, counseling, etc.) - Physical: if intervention is exercise/rehabilitation
- I_other: any other intervention type
- Control: any comparison intervention

Rules

- Identify type of intervention clearly
- Use medical standard terminology (UMLS/SNOMED-like language) to normalise the values
- Dosage should include numerical value for its respective drug - Do NOT add new variables
- Do NOT remove keys
- If no update, return same JSON
- No explanation text

V. Zero-shot instruction tuned prompt to parse full-text for Outcome component

You are a biomedical information extraction expert working on clinical research papers. Your task is to VERIFY and REFINE previously extracted outcome variable and divide it into 5 variables(Mortality, Adverse event, life impact, clinical, O_other) related to outcome elements of PICO framework by analysing each input chunk of the Results section.

Update only 5 Variables:

- Mortality: if any overall death/survival
- Adverse_Event: if any unintended consequences
- Life_Impact: if intervention affects the physical, social, cognitive and emotional

function, delivery of care and health status

- Clinical: it measures any clinical/physiological signs, symptoms
- Numeric: statistical measure values such as p-value, mean, standard deviation
- O_other: Any other outcome such as cost, hospital stays or any other resource related

Rules

- Do NOT add new variables
- For each outcome type, return its relevant numeric values (mean, standard deviation, p-values)
- Do NOT remove keys
- If no update, return same JSON
- No explanation text
- Use medical standard terminology (UMLS/SNOMED-like language)